

Infinite Length, Finite Volume

Two arches, one shared ball, and two sums that disagree

Name _____

Date _____

Mathemalchemy is a travelling mathematical art installation built by a team of 24 artists and mathematicians (mathemalchemy.org). Two of the things in it are infinite series you can walk up to and look at: a tortoise's tiled path, and a pair of arches made of embroidered thread balls. This worksheet works out the mathematics of both.

The surprise is waiting in Part 4. One of the two arches is *infinitely long* – laid end to end its balls would never stop – and yet all of those infinitely many balls together hold only a finite amount of material. Parts 1 through 3 build the tools to show it.

You will need a calculator. Nothing here needs a ruler or a measurement: every length is given as a multiple of the first ball's diameter, so the first ball is 1 unit across no matter how big it really is.

Part 1. Zeno's Path: a warm-up you already believe

Tess the Tortoise walks a tiled path to Integral Hill. The path is laid in sections, and each section is half as long as the one before it.

Zeno's dichotomy paradox says Tess can never arrive: to reach the end she must first reach the halfway point, then half of what remains, then half of *that*, and so on forever. Infinitely many tasks, each waiting on the one before. The path in the gallery is built to match – the sections really do halve.

Measure everything as a fraction of the whole path.

1.1 Fill in the table. “Total so far” is the running sum of the sections walked.

section n	length of section n	total walked so far, S_n
1	$1/2$	$1/2$
2	$1/4$	
3		
4		
5		

1.2 Look down the “total so far” column. Each entry is 1 minus something. Write S_n in closed form: $S_n =$

1.3 Now prove it, rather than spotting it. Write out $S_n = \frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{2^n}$, multiply the whole thing by $\frac{1}{2}$, and subtract the second line from the first. Almost everything cancels.

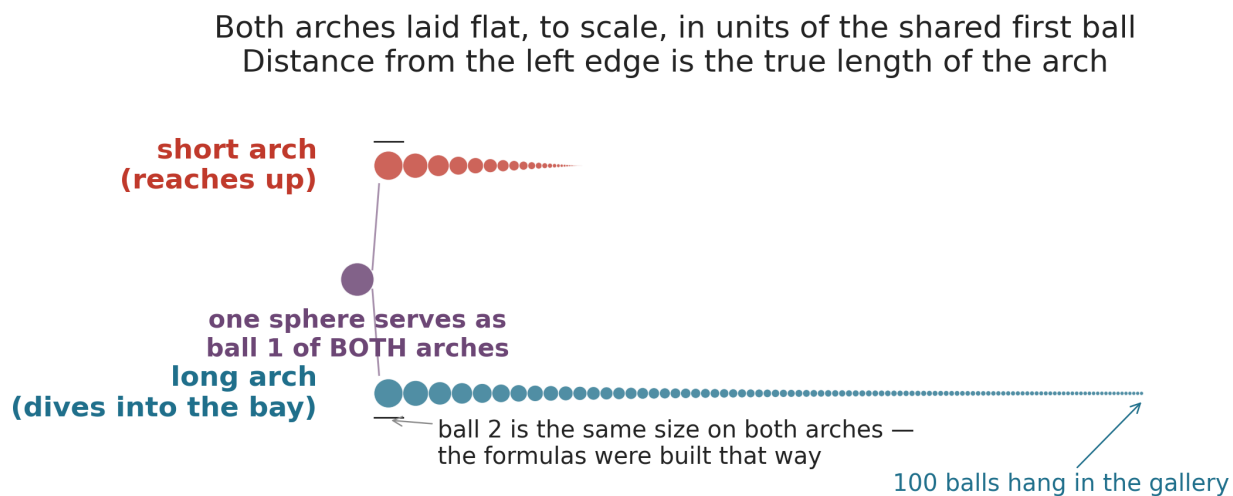
1.4 What is $\lim_{n \rightarrow \infty} S_n$?

1.5 After Tess finishes section n , how much of the path is still ahead of her? Compare that with the length of the section she has just walked.

1.6 In a sentence: the path is made of infinitely many sections, but it has a finite length. Explain how both can be true.

Part 2. The short arch: a geometric series in disguise

Two arches of thread-wrapped balls rise from one shared first ball. The short arch reaches up; the long arch dives into the bay. On both, the balls shrink toward nothing.



The makers gave each arch a formula. The short arch's ball diameters are

$$d_n = d_1 \left(\frac{4}{5} \right)^{\frac{2(n-1)}{3}}, \quad n = 1, 2, 3, \dots$$

where d_1 is the diameter of the shared first ball. Take $d_1 = 1$ unit throughout.

2.1 Compute the first three diameters, exactly and to four decimal places.

2.2 Is this a geometric sequence? Do not just say yes – show that the ratio d_{n+1}/d_n is the same number for *every* n . (Subtract the exponents.)

2.3 Fill in the first six balls. Round to four decimal places.

ball n	diameter d_n	running total S_n
1	1.0000	1.0000
2	0.8618	
3		
4		
5		
6		

Check yourself: at $n = 4$ the exponent is $2(3)/3 = 2$, so d_4 should be exactly $(4/5)^2 = 0.64$. The same happens at $n = 7$ and $n = 10$, where d_n is exactly $(4/5)^4$ and $(4/5)^6$.

2.4 Now add up *all* of them – infinitely many balls. Use the geometric series formula and the ratio from 2.2.

2.5 Here is the part that makes convergence feel real. The balls after ball N are themselves a geometric series, so

$$S_\infty - S_N = \frac{d_1 r^N}{1 - r}.$$

Divide that by S_∞ from 2.4. What is the *fraction* of the arch still to come after ball N ? Then evaluate it at $N = 20$ and at $N = 60$.

2.6 Complete the sentence: the short arch is first-ball diameters long, and no number of extra balls will ever push it past that.

Part 3. The long arch: where the intuition breaks

Same construction, same shrinking balls, a different formula.

The long arch's diameters are

$$D_n = D_1 \left(\frac{4}{n+3} \right)^{2/3}, \quad n = 1, 2, 3, \dots$$

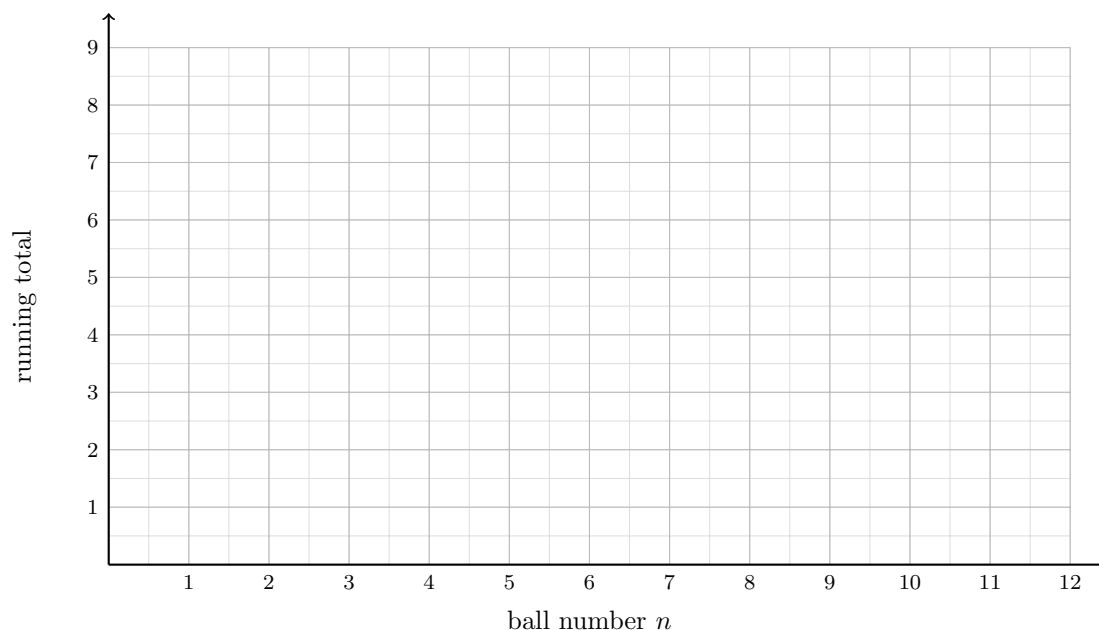
and $D_1 = d_1 = 1$: the first ball is one physical sphere belonging to both arches.

3.1 Check the two arches really do start together. Compute D_1 and D_2 from the formula above and compare with d_1 and d_2 from Part 2.

3.2 Fill in the two running-total columns. The diameters are given, so this is addition only. Round to four decimals.

ball n	short arch		long arch	
	d_n	running total S_n	D_n	running total L_n
1	1.0000	1.0000	1.0000	1.0000
2	0.8618	1.8618	0.8618	1.8618
3	0.7427		0.7631	
4	0.6400		0.6886	
5	0.5515		0.6300	
6	0.4753		0.5824	
7	0.4096		0.5429	
8	0.3530		0.5095	
9	0.3042		0.4808	
10	0.2621		0.4558	
11	0.2259		0.4338	
12	0.1947		0.4143	

3.3 Plot both running totals against n on the grid below – two curves on the same axes. Then draw a horizontal dashed line at the short arch’s infinite total from 2.4.



3.4 Both arches have balls shrinking toward zero: $\lim_{n \rightarrow \infty} d_n = \lim_{n \rightarrow \infty} D_n = 0$. A student says: “the terms go to zero, so both sums are finite.” What is wrong with that?

3.5 Try the ratio test on each arch. For the long arch, simplify D_{n+1}/D_n first, then evaluate it at $n = 1, 10, 100, 1000$ and take the limit.

3.6 Something else has to settle the long arch. Substitute $k = n + 3$ to rewrite $\sum D_n$ as a constant times a p -series, identify p , and conclude.

3.7 That rewriting threw away the terms $k = 1, 2, 3$. Why can that not change the answer?

3.8 Read this off your table in 3.2. At which ball does the long arch's running total pass the short arch's *entire infinite total* – your answer to 2.4, everything the short arch will ever achieve?

3.9 (Harder.) The integral test does more than decide convergence: it estimates the sum. Using $\int x^{-2/3} dx = 3x^{1/3}$, the first N balls of the long arch come to roughly

$$L_N \approx 3 \cdot 4^{2/3} \left[(N+3)^{1/3} - 4^{1/3} \right].$$

(a) About how many balls would it take to reach a length of 50 first-ball diameters? (b) The arch is currently 100 balls long. Roughly how many balls would it take to make it *twice* as long? What feature of the formula tells you that?

Part 4. Now weigh them

The long arch never ends. So how much stuff is in it?

4.1 Commit to a guess before you compute anything. Circle one in each row.

The total volume of all the long arch's spheres is. . .	infinite	finite
The arch holding more total volume is. . .	the short one	the long one

4.2 A sphere of diameter D has radius $D/2$. Write its volume in terms of D .

4.3 So volume depends on the *cube* of the diameter. Cube the long arch's diameter formula and simplify. Watch what happens to the exponent.

4.4 Write $\sum V_n$ as a constant times a p -series. What is p now, and what does that tell you?

4.5 Find that total exactly. You may use the Basel sum $\sum_{k=1}^{\infty} \frac{1}{k^2} = \frac{\pi^2}{6}$ – but the arch’s series starts at $k = 4$, so subtract what is missing.

4.6 Do the same for the short arch. Its diameters are geometric with ratio $(4/5)^{2/3}$, so its volumes are geometric too. What is the volume ratio, and what is the total?

4.7 Fill in the summary. This is the whole worksheet in four boxes.

	sum of the diameters	sum of the volumes
short arch		
long arch		

4.8 How many times more volume does the endless arch hold than the one that stops?

4.9 Explain to someone who has never taken calculus how an arch can go on forever and still be made out of a finite amount of material.

Note. What we have computed is the sum of the volumes of infinitely many idealized spheres, whose diameters decrease according to the given formulas. We are not making any claim about the material used to make the spheres.

Part 5. Why two-thirds?

The exponent $2/3$ looks arbitrary. It is not. Here is what it has to accomplish.

Suppose the makers had used a general exponent p instead:

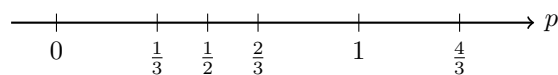
$$D_n = D_1 \left(\frac{4}{n+3} \right)^p.$$

The two conditions we are putting on p are exactly the two results of Parts 3 and 4: the series of *diameters* must diverge, and the series of *volumes* must converge. Which values of p do both?

5.1 For which values of p does the total *length* $\sum D_n$ diverge?

5.2 For which values of p is the total *volume* $\sum V_n$ finite? (Remember volume cubes the diameter.)

5.3 Put both conditions together and mark the allowed range on the number line. Then mark where the real arch sits.



Write the allowed range as an inequality:

5.4 Discussion. The harmonic choice $p = 1$ is also inside the window, and it is the more obvious pick for a mathematician. Compute the hundredth ball's diameter, D_{100}/D_1 , for $p = \frac{2}{3}$ and for $p = 1$. Why might a maker of thread balls have preferred two-thirds?

Part 6. The same paradox, in the same exhibit

Nearby, paper snowflakes fall among the pages of a cavalcade. They are Koch snowflakes, and they do the same trick in a different costume.

A Koch snowflake starts from an equilateral triangle of area A . At each stage, every edge is replaced by four edges one third as long, with a new triangular bump pushed out in the middle.

6.1 What happens to the *perimeter* at each stage? What is it after k stages, and what happens as $k \rightarrow \infty$?

6.2 Now the area. At stage k the snowflake gains $3 \cdot 4^{k-1}$ new triangles, each of area $A/9^k$. Sum the added areas and add the original triangle.

6.3 In one sentence, what do the Koch snowflake and the long arch have in common?

Back where we started. Tess's path in Part 1 is infinitely many sections of finite total length, and it convinced nobody that anything strange was happening. The long arch is infinitely many balls of *infinite* total length and finite total volume. Same kind of object, same kind of sum, and the answer depends entirely on how fast the pieces shrink and on what you choose to measure.